

The Ultimate NLP Terminology Guide

1. The Basics (Foundation)

These are the building blocks. You cannot do NLP without these.

Tokenization

- Definition: The process of breaking down a text string into smaller pieces called tokens (words, subwords, or characters).
- Example: The sentence "I love AI" becomes ["I", "love", "AI"].
- Use Case: The first step for any NLP model. It translates raw text into a format that a computer can process numerically.

Vocabulary

- Definition: The unique set of tokens that your model knows.
- Example: If your dataset has the words "Hello" and "World", your vocabulary size is 2 (assuming no repetition).
- Use Case: Determines the size of the embedding matrix. If a word isn't in the vocabulary (OOV), the model can't recognize it.

Corpus (Plural: Corpora)

- Definition: A large, structured set of texts used for training.
- Example: The "Brown Corpus" or "Wikipedia Dump".

Data Augmentation

- Artificially increasing the size of the dataset by modifying existing text.
- Replacing "great" with "amazing" or back-translating (English→French→English).

Morphology

- The study of the structure and form of words (roots, prefixes, suffixes).
- "Untouchable" = "Un" (prefix) + "touch" (root) + "able" (suffix).

Semantics

- The study of *meaning*. What do the words actually refer to?
- Recognizing that "Bank" means a river side vs. a financial institution.

Pragmatics

- Understanding meaning based on *context* and real-world knowledge.
- If someone says "It's cold in here," the pragmatic meaning is "Please close the window."

Stop Words

- Definition: Commonly used words (like "the", "is", "in") that are often filtered out because they add little meaning.
- Example: "*The cat is on the mat*" becomes ["cat", "mat"].
- Use Case: Improves search engine efficiency. Warning: Usually skipped in modern LLMs (like GPT) because they rely on context/position.

Stemming & Lemmatization

- Definition: Reducing words to their root form.
 - Stemming: Cuts off the end aggressively (e.g., "running" → "run", "studies" → "studi").
 - Lemmatization: Uses a dictionary to return a valid base word (e.g., "studies" → "study").
- Use Case: Standardizing text. E.g., searching for "walking" should return docs containing "walked".

N-grams

- Definition: Contiguous sequences of N words.
 - Example:
 - *Unigram (1)*: ["I", "love", "NLP"]
 - *Bigram (2)*: ["I love", "love NLP"]
 - Use Case: Traditional language modeling (for predicting the next word) and spam detection (e.g., looking for "free money" as a bigram).
-

2. Vectors & Understanding (Numerical Representation)

How we turn words into math.

Bag of Words (BoW)

- Definition: A sparse vector representing the frequency of words in a document, ignoring grammar and order.
- Example: "Cat eats mouse" = [1, 1, 1]; "Mouse eats cheese" = [0, 1, 1] (depending on vocab).
- Use Case: Document classification (e.g., is this email about sports or politics?).

TF-IDF (Term Frequency-Inverse Document Frequency)

- Definition: Weighs how important a word is to a document in a collection.
 - High score = Frequent in this doc, but rare in other docs.
- Use Case: Search engine ranking. If you search "NLP", a doc that mentions it 10 times ranks higher than a doc that mentions it once.

Word Embeddings (Word2Vec / GloVe)

- Definition: Dense vector representations (e.g., arrays of 100 floating numbers) that capture *semantic meaning*.
- Key Rule: King - Man + Woman
- $\approx$ Queen.

- Use Case: The backbone of early deep learning NLP tasks. Allows models to understand that "cat" and "kitten" are similar.
-

3. Syntax & Structure (Grammar)

How words relate to each other.

Part-of-Speech (POS) Tagging

- Definition: Identifying the grammatical role of a word (Noun, Verb, Adjective, etc.).
- Example: "AI is fun"
- → AI(Noun), is(Verb), fun(Adjective).
- Use Case: Text-to-Speech (intonation changes depending if a word is a verb or a noun).

Named Entity Recognition (NER)

- Definition: Extracting real-world entities from text (People, Locations, Organizations, Dates).
- Example: "Elon Musk lives in Texas"
- → Person: Elon Musk, Location: Texas.
- Use Case: Automatically tagging articles, creating knowledge graphs, or automatic data redaction (hiding sensitive PII).

Dependency Parsing

- Definition: Analyzing the grammatical structure to find relationships between words (e.g., Subject-Verb-Object).
 - Use Case: Question Answering (finding the subject to identify what the question is about).
-

4. Advanced Concepts (Model Architecture)

The 'How' of modern AI.

Attention Mechanism

- Definition: A technique that allows a model to focus on the most relevant parts of the input sequence while generating an output.
- Example: Translating "It is hot" to French. When generating the word for "hot", the model pays attention specifically to "hot" in the source.
- Use Case: Machine Translation. It solves the "bottleneck" issue of older RNNs.

Transformers

- Definition: The architecture that revolutionized NLP. It relies solely on Attention (and no RNNs/CNNs).
- Use Case: The engine behind GPT, BERT, and Llama. Used for everything from text generation to code writing.

BERT (Bidirectional Encoder Representations from Transformers)

- Definition: An encoder-only model. It reads text bidirectionally (left-to-right AND right-to-left).
- Use Case: Best for understanding tasks—Sentiment Analysis, Text Classification, and Question Answering.

GPT (Generative Pre-trained Transformer)

- Definition: A decoder-only model. It reads text unidirectionally (left-to-right) and predicts the next token.
 - Use Case: Best for generative tasks—Story writing, code completion, and chatting (ChatGPT).
-

5. Evaluation Metrics (Did it work?)

How we measure success.

Perplexity

- Definition: A measure of how "surprised" a language model is by a test set. Lower Perplexity = Better model.
- Use Case: Comparing two language models to see which writes more fluent text.

BLEU Score (Bilingual Evaluation Understudy)

- Definition: Compares machine-generated text to human reference text. Measures precision (how many n-grams match).
- Use Case: Machine Translation. Scoring how close a translation is to a professional human translation.

ROUGE Score (Recall-Oriented Understudy)

- Definition: Measures Recall (how much of the reference text is captured).
 - Use Case: Summarization. Checking if the summary contains all the key facts from the original text.
-

6. The "Gotchas" (Problems in NLP)

Out-of-Vocabulary (OOV)

- Definition: When the model encounters a token it has never seen during training.
- Solution: Subword tokenization (e.g., BPE - Byte Pair Encoding). Example: "unhappy" is split into "un" + "happ" + "y" so it can guess the meaning.

Bias in AI

- Definition: The model inherits stereotypes or prejudices present in the training data.
- Use Case: Important for fairness. If a model always associates "Nurse" with "Women", that's bias.

Hallucination

- Definition: When an LLM generates content that is factually incorrect or nonsensical, but sounds confident.
- Example: An AI creates a news article about an event that never happened.

CLASSICAL ML & STATISTICAL NLP

Before Transformers, these were state-of-the-art.

Term	Definition	Example	Use Case
Naive Bayes	A probabilistic classifier based on Bayes' Theorem, assuming all features are independent.	Calculating: $P(\text{Spam} \mid \text{"Free Money"})$ vs $P(\text{Not Spam} \mid \text{"Free Money"})$.	Use Case: Fast spam email filtering.
Support Vector Machine (SVM)	A classifier that finds the best "hyperplane" to separate different classes of data.	Separating "Positive" reviews from "Negative" reviews in a high-dimensional space.	Use Case: Text Categorization (News topics).

Hidden Markov Model (HMM)	A statistical model where the system switches between hidden states over time.	States = [Noun, Verb, Noun]; Observations = ["I", "love", "NLP"].
---------------------------	--	--

DEEP LEARNING ARCHITECTURES (The Evolution)

The neural networks that paved the way for ChatGPT.

Term	Definition	Example	Use Case
Recurrent Neural Network (RNN)	A network with loops, allowing information to persist across sequences.	Processes text word-by-word, maintaining a "memory" of previous words.	Early text generation and translation.
Long Short-Term Memory (LSTM)	A specialized RNN that can remember information for longer periods (solves the vanishing gradient problem).	Can remember the subject of a sentence even after 50 words.	Used in Google Translate before Transformers.

Gated Recurrent Unit (GRU)	A simpler, faster version of LSTM (fewer gates).	Similar to LSTM but uses less memory.	Used when you have less compute power.
Vanishing / Exploding Gradients	During training, the "error" signal gets smaller (vanishes) or larger (explodes) as it travels back through layers.	A 100-layer RNN can't learn the first word's effect because the gradient is 0.	The problem that Attention mechanisms solved.
Backpropagation Through Time (BPTT)	The specific way RNNs calculate errors by unrolling the network over time steps.	Treating "I", "Love", "AI" as three layers in time.	Training RNNs, LSTMs, and GRUs.
Sequence-to-Sequence (Seq2Seq)	An architecture with an Encoder (reads input) and a Decoder (generates output).	Encoder reads English "Hello"; Decoder generates French "Bonjour".	